Discusssion of Broxterman, Larson, Yezer “Characteristics of a Sufficient Statistic to Measure City Housing Prices”

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Contributions

- Evidence of heterogeneous price growth across space
  - Builds on e.g. Landvoigt, Piazessi, + Schneider
- Offers a new methodology for aggregating to the metro level
- Makes us think about counting units vs hedonic quality aggregation
  - Highly salient to the "shoebox condo" debate

Target for Laspeyre's index

$$\frac{\Delta r_t^*}{r_{t-1}} = \frac{\sum_k r_t(k) h_{t-1}(k)/H_{t-1}(k)}{\sum_k r_{t-1}(k) h_{t-1}(k)/H_{t-1}(k)}.$$

- r price or rent, h individual home, H sum of housing units
- What is an h an important issue emphasized by authors
- Instinct: units better but imperfect
  - Not easy to see common use where luxury homes get way more weight
  - Tiny condos getting much cheaper in YYZ helpful?
- Observation: weights essentially an indicator for being transacted
  - More evidence on non-representativeness of that sample
    - Intuition for channel
    - New homes a good example (excluded!)

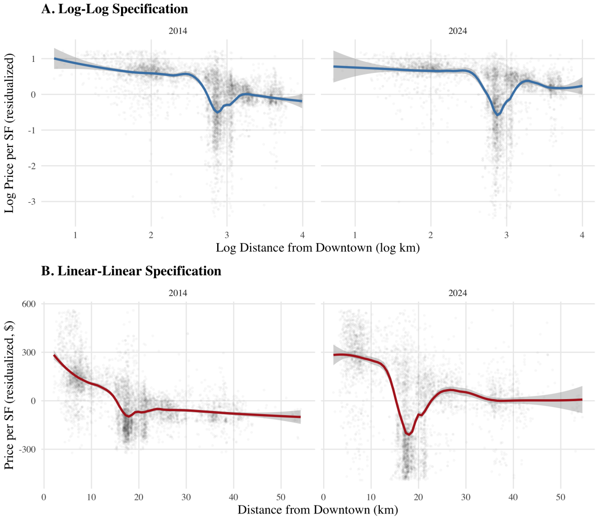
Emphasis on location as heterogeneity

- Result from familiar monocentric city model
  - $\frac{dr(k)}{dk} = -\frac{\frac{d\text{commute cost}}{dk}}{h_t}$
  - If I move a bit further, $|h\dot{r}| = |\dot{\text{commute cost}}|$
- Also if constant value (construction + option premium) at boundary
  - $d\left[r(k) - r(b(t))\right] = \theta\dot{b} = dr(k) = \text{constant}$
  - Then need constant value in distance
  - Some condition on income and price elasticities
- But linear commute cost ??
  - 41st to 61st St in Manhattan vs random mile in the suburbs
- Suppose $\ln p(h,k) = \ln h + \theta \log \frac{b}{k}$, then $g'(k) = 0$ (with light conditions.)

Work plan as expected

- Show clearly log beats linear price gradient in Vancouver
- Explain that location not the logical heterogenity to focus on
  - Or at least that this is overblown "Unless highly restrictive preference assumptions are imposed, appreciation rates cannot be spatially invariant."
- But not exactly what happened . . .

Vancouver condo sales by year: distance gradients



Vancouver Condo: fit

- Log-log fits better 2014
- Subsequent years
  - Gradient flattens a lot
  - Action is at outer edge
  - Levels gets to be much better fit
- So $g'(k) \neq 0$ as authors emphasize
- Reg level P on log distance generally fits by far best
  - inverse of old literature
  - but doesn't capture flattening
  - Old literature does get flattening right
  - True for single and condo

Pairwise correlations among Vancouver vs Suburban types

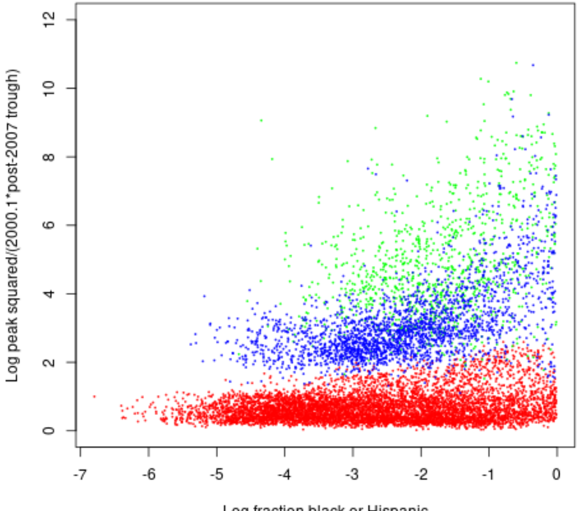
Type matters more than location?

	VanSingle	VanCondo	FVSingle	FVCondo
VanSingle	1.00	0.71	0.82	0.54
VanCondo	0.71	1.00	0.71	0.70
FVSingle	0.82	0.71	1.00	0.71
FVCondo	0.54	0.70	0.71	1.00

- Log index growth 2014-2024: FV condo: 10.7%
- GV single: 5.9%
- Story: "flight from quality" after foreign buyer ban?

GFC: heterogeneity driven by subprime propensity
Zip code price movements boom bust (source, Zillow)

Red: Flyover, Blue: Coastal, green: Sand State



Transaction weight: spatial smoothing vs tract indexes?

- Index = sum over tracts tract weight times tract price change
- Start from equal tract weights
- Then transaction weight = 1/number of transactions in tract
  - This seems like an invitation to outlier problems
- Suggestion: borrow from kernel estimation literature
- This can extend to multiple dimensions with cell weights
  - Price point (target income/credit constraints)
  - Investor/owner occupier (condo/single)
  - ...

What is the question indexes answer?

- Ask a realtor (or self) what is my home worth?
  - Don't ask initial purchase price
  - Unlikely mention index growth
  - Don't discuss other property types
- Mark-to-market LTV distribution?
  - But then middle of distribution unlikely the object of interest
- What decision-maker needs to aggregate to 1-type-metro-level?
  - Portfolio efficiency for investor? X
  - Workout decision for lenders? X
  - Government transfer tax forecast? $\surd$
- Has anyone ever messed up an optimization with wrong index info?

Is AHS price data bananas?

- 2015-2019 Table 1 shows Atlanta 18% vs 13% per year appreciation
- FHFA per FRED says about 7%
- Washington, D.C. Table 1 says about 2%
- FHFA per FRED says about 3.7%.
- Aggregate Zillow zip indexes by metro central/suburbs?
- Strong suggestion: use better data and see if differences as large

Summary

- Important takeaways from paper
  - Price growth not constant across submarkets
  - Spatial differences in growth stand out
  - Aggregation weights depend on target unit of measure
- Suggestions
  - Consider whether spatial differentiation warrants focus
  - Sane models *can* imply constant appreciation in distance
    - My evidence mixed and weird on gradient
    - Discuss this literature (Mills, McMillen, Coulson, ...) more
  - Use better data and work with kernel smoothing techniques
    - Inverse weights likely problematic
  - Do more work to convince that large aggregates have value